Assignment 2: Email Spam Classification

Dataset: Email Spam Classification Dataset (Kaggle)

Binary classification (0 = Not Spam, 1 = Spam) using K-Nearest Neighbors and Support Vector Machine, with a performance comparison.

In [1]:
```
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn.neighbors import KNeighborsClassifier
from sklearn.svm import SVC
from sklearn.metrics import accuracy_score, classification_report,
confusion_matrix, ConfusionMatrixDisplay
import matplotlib.pyplot as plt
```

Load the dataset

In [2]:
```
df = pd.read_csv("emails.csv")
df = df.drop(columns=[c for c in df.columns if c.lower().startswith("email
no")], errors="ignore")
df.head()
```

Out[2]:

	the	to	ect	and	for	of	a	you	hou	in	...	connevey	jay	valued	lay	infr
0	0	0	1	0	0	0	2	0	0	0	...	0	0	0	0	
1	8	13	24	6	6	2	102	1	27	18	...	0	0	0	0	
2	0	0	1	0	0	0	8	0	0	4	...	0	0	0	0	
3	0	5	22	0	5	1	51	2	10	1	...	0	0	0	0	
4	7	6	17	1	5	2	57	0	9	3	...	0	0	0	0	

5 rows × 3001 columns

In [3]:
```
print(df.shape)
df["Prediction"].value_counts()
```

```
(5172, 3001)
```

Out[3]:
```
Prediction
0    3672
1    1500
Name: count, dtype: int64
```

Train/test split

In [4]:
```
X = df.drop(columns=["Prediction"])
y = df["Prediction"]

X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2,
random_state=42, stratify=y)
```

Scale features

In [5]:
```
scaler = StandardScaler()
X_train_scaled = scaler.fit_transform(X_train)
X_test_scaled = scaler.transform(X_test)
```

K-Nearest Neighbors

In [6]:
```
knn = KNeighborsClassifier(n_neighbors=5).fit(X_train_scaled, y_train)
knn_preds = knn.predict(X_test_scaled)
knn_acc = accuracy_score(y_test, knn_preds)

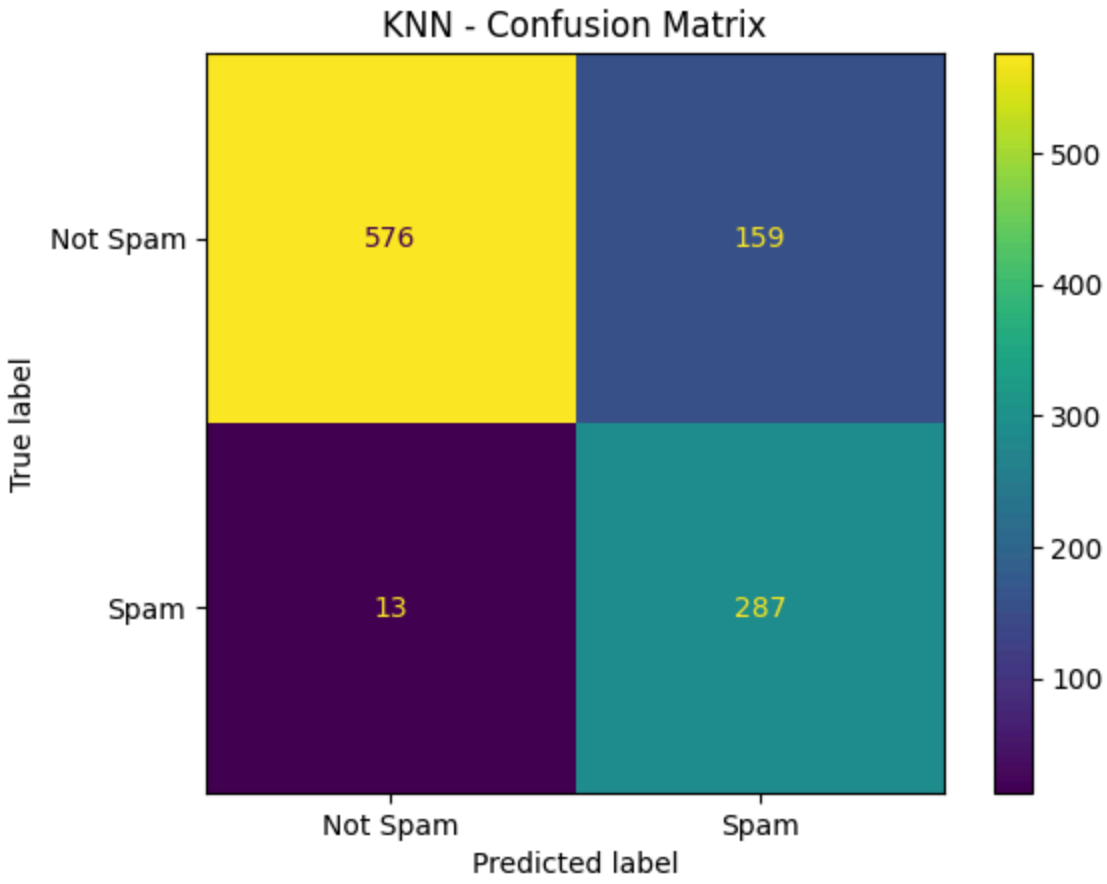
print(f"KNN accuracy: {knn_acc:.4f}\n")
print(classification_report(y_test, knn_preds, target_names=["Not Spam",
"Spam"]))
ConfusionMatrixDisplay(confusion_matrix(y_test, knn_preds), display_labels=
["Not Spam", "Spam"]).plot()
plt.title("KNN - Confusion Matrix")
plt.show()
```

```
KNN accuracy: 0.8338

              precision    recall  f1-score   support

    Not Spam       0.98      0.78      0.87       735
        Spam       0.64      0.96      0.77       300

    accuracy                           0.83      1035
   macro avg       0.81      0.87      0.82      1035
weighted avg       0.88      0.83      0.84      1035
```



Support Vector Machine

In [7]:
```
svm = SVC(kernel="rbf", random_state=42).fit(X_train_scaled, y_train)
svm_preds = svm.predict(X_test_scaled)
svm_acc = accuracy_score(y_test, svm_preds)

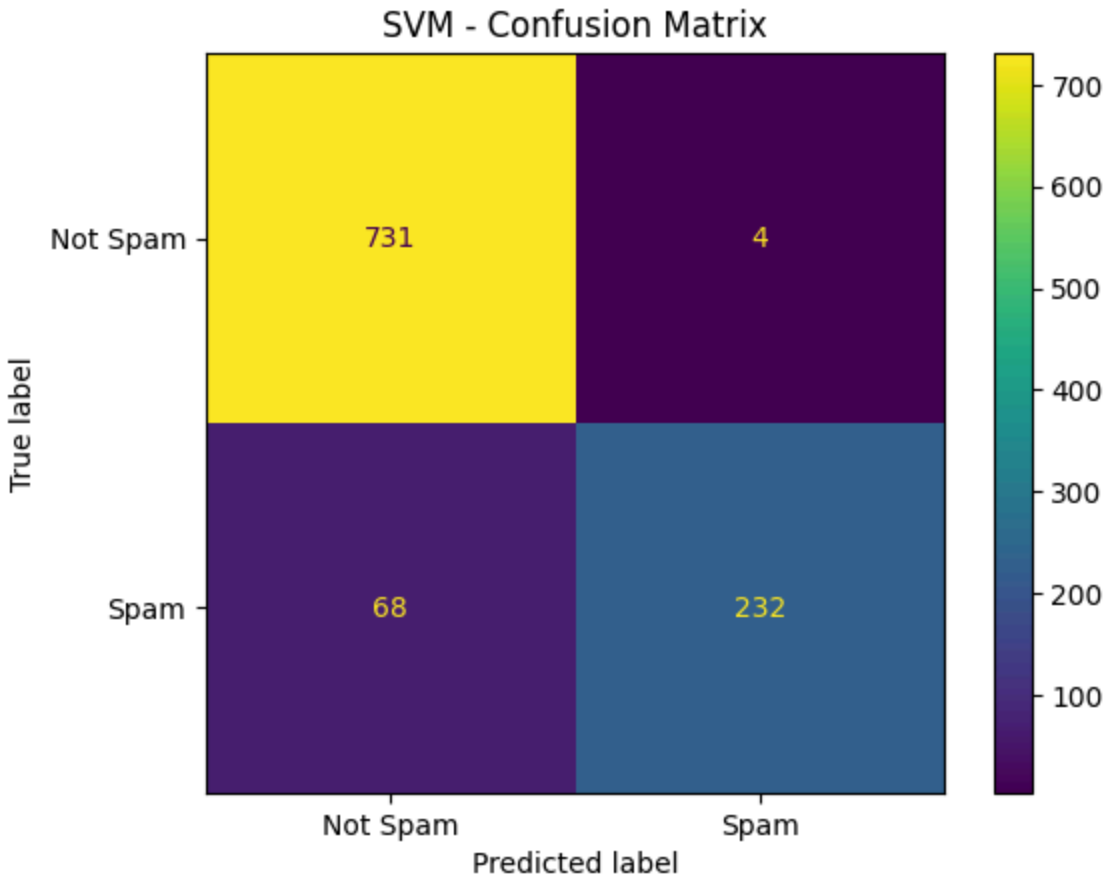
print(f"SVM accuracy: {svm_acc:.4f}\n")
print(classification_report(y_test, svm_preds, target_names=["Not Spam",
"Spam"]))
ConfusionMatrixDisplay(confusion_matrix(y_test, svm_preds), display_labels=
["Not Spam", "Spam"]).plot()
plt.title("SVM - Confusion Matrix")
plt.show()
```

```
SVM accuracy: 0.9304

              precision    recall  f1-score   support

    Not Spam       0.91      0.99      0.95       735
        Spam       0.98      0.77      0.87       300

    accuracy                           0.93      1035
   macro avg       0.95      0.88      0.91      1035
weighted avg       0.93      0.93      0.93      1035
```



Performance comparison

In [8]:
```
pd.DataFrame([
    {"Model": "KNN", "Accuracy": knn_acc},
    {"Model": "SVM", "Accuracy": svm_acc},
])
```

Out[8]:

	Model	Accuracy
0	KNN	0.833816
1	SVM	0.930435